SPINE Volume 27, Number 5, pp 453–459
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Morphometric Study of the Lumbar Pedicle in the Indian Population as Related to Pedicular Screw Fixation

Sajal R. Mitra, MS, Sandeep P. Datir, MS, DNB, and Sanjay O. Jadhav, MS

Study Design. A morphometric study of lumbar vertebral pedicular parameters in cadavers and comparison with previous studies in the literature was conducted.

Objectives. To suggest dimensions for pedicular implants in the Indian population, and to improve the pedicular screw placement technique.

Summary of Background Data. Detailed knowledge of pedicle morphometry is critical for proper placement of a transpedicular screw. The size and shape of the vertebral pedicle vary between different races. Morphometric studies have been conducted in white and nonwhite populations (*e.g.,* Chinese, Koreans).

Methods. The vertebral pedicles at L1–L5 were studied in 20 cadavers by direct roentgenographic and computerized tomographic scan methods. The following parameters were studied: transverse diameter, transverse angle, sagittal diameter, sagittal angle, chord length, interpedicular distance, medial and lateral cortical thickness.

Results. Transverse diameter was largest at L5 (16.19 mm) and smallest at L1 (7.05 mm). The transverse angle was largest at L5 (29°) and smallest at L1 (9°). The pedicles were directed cranially in the sagittal plane at all lumber levels except L5. The sagittal angle was largest at L5 (29°) and smallest at L1 (9°). Chord length was largest at L2 (47.5 mm) and smallest at L1 (46.01 mm). The values of linear measurements were smaller in females at all levels.

Conclusions. On the basis of this limited study in a subset of the Indian population, it appears that the transverse diameter and pedicle entrance point differ from those in the white population. The results suggest that a 5-mm screw would be safer in the upper lumbar levels (L1, L2), and 6-mm screw in the lower lumbar levels (L3–L5). The pedicle entrance point migrates laterally for lower lumbar levels, especially at L5. The medial pedicle cortex can be safely sounded while the pedicle is probed. [Key words: lumbar, morphology, pedicles, spinal fixation, spine] **Spine 2002;27:453–459**

Posterior spinal instrumentation with the help of rods, hooks, and wires is a time-honored method of spinal fixation. However, absence of the lamina because of trauma or laminectomy precludes the use of these systems. Moreover, multiple vertebral levels must be spanned when the Harrington and Luque systems are used because the three-dimensional rigid link between implant and vertebral body is lacking.[2,9,14,26] This leads to substantial decrease in mobility of the spine,[26] which is more relevant in the lumbar region. The Harrington instrumentation system requires at least one intact soft tissue column (*i.e.*, an anterior longitudinal ligament).[5]

Because of these factors, transpedicular screw fixation methods are increasingly used worldwide.[6] Pedicular fixation is more stable and versatile because it provides three-dimensional fixation. Also it has been proved that the pedicle is the strongest part of the vertebra even in osteoporotic bone.[12,19]

Detailed knowledge of pedicle morphometry is critical for proper placement of the transpedicular screw and to avoid inadvertent penetration of the pedicular wall. Studies already have been conducted in white and a few nonwhite populations (Chinese and Korean).[1,3,4,7,8,10,13,16–18,22,24,27] However, no such study has been conducted in an Indian population.

Many studies have not reported all the desired morphometric dimensions related to pedicular screw placement. In this report, an attempt is made to compare and analyze the parameters in both the current study and the studies found in the literature.

Methods

For this study, 20 cadavers (18 males and 2 females) were selected from the Department of Anatomy, Government Medical College and Hospital, Nagpur, India. The cadaver ages ranged from 20 to 70 years for the male cadavers. The two female cadavers were 55 and 60 years of age respectively. The cadavers showing obvious deformity in the spine or a fractured spine were excluded from this study.

The cadavers were dissected using the midline posterior approach from T12 to sacrum. The facet joints and the vertical bony crest just below it were cleared of the soft tissue, and the segment of spine from L1 to sacrum was removed.

All the diameters and distances were measured using a Vernier caliper (resolution, 0.1 mm). All the angular measurements were made with a goniometer (resolution, 1°) on lateral roentgenograms and directly on the spinal column.

Direct measurements were taken first on the intact vertebral column including the interpedicular distance, the sagittal diameter, and the sagittal angle (Figure 2) from L1 to L5. Roentgenograms of the vertebral column were taken including the sagittal diameter, the sagittal angle, and the interpedicular distance.[18,21,27] The cassette was kept 100 cm from the tube (≈33 inches), and a metal measuring guide was used to determine the magnifying factor, which was found to be 0. The sagittal angle was measured by drawing one line along the axis of the pedicle and another line along the superior border of vertebral body, as described by Olsewski et al.[18]

From the Department of Orthopaedics, Government Medical College, Nagpur, India.
Acknowledgment date: February 3, 2000.
First revision date: April 14, 2001.
Acceptance date: August 13, 2001.
The manuscript submitted does not contain information about medical device(s).
No funds were received in support of this study.

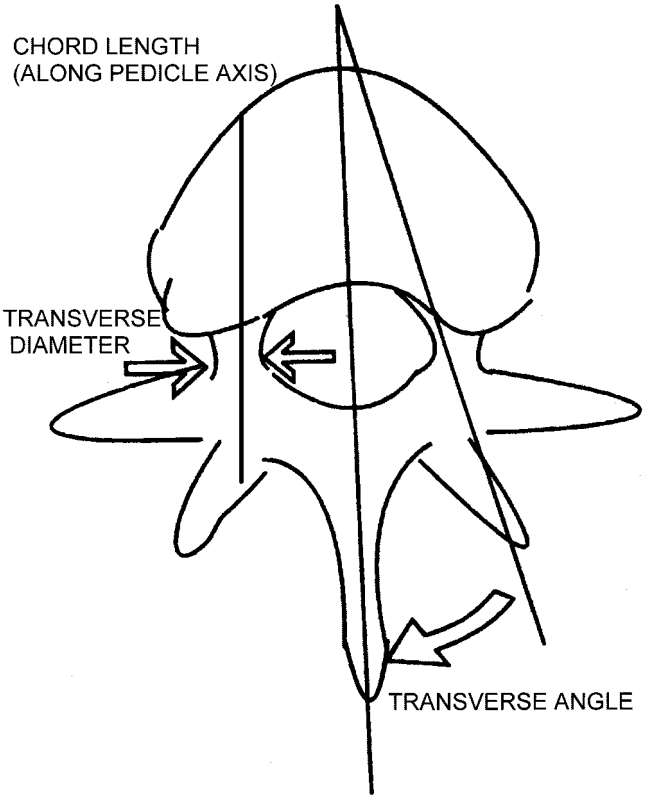


Figure 1. The method for measuring the transverse diameter and the transverse angle.

A computerized roentgenogram was performed, and 5-mm-thick slices (midpedicle cut) were cut at the pedicle isthmus in the axial plane, which was determined from the sagittal view. Measurements were taken directly from the hard copy films using computer software for distance measurement. Angular measurements were taken with the help of a goniometer directly from the computed tomography (CT) scan films. The following parameters were studied on the CT scan: transverse diameter, transverse angle, chord length, medial and lateral cortical thickness.[10]

Next, vertebrae were cut at the midpedicle level using an oscillating fret hacksaw, and the following parameters were studied directly: transverse diameter, transverse angle (Figure 1), chord length, and pedicle entrance point.[7,19,20,25]

Pedicle Entrance Point. The segment of the spine from L1 to L5 was studied visually to determine the pedicle entrance point. As stated by Roycamille et al,[19,20] this point lies over a typical bony crest hidden by soft tissue 1 mm below the facet joint. Magrel[15] described the entry point as being where the lateral border of the superior facet intersects the mid–transverse process. Weinstein et al[25] advocated a lateral approach to the pedicle, and described the entry point as located at the "nape of the neck" of the superior articular facet.

In the current study, the method described by Hou et al[7] was used. Visual examination in the lateral plane allowed determination of the pedicle center. It was noted whether this point lay at the top, upper half, midpoint, or lower half of the transverse process. The superior facet of the vertebra was divided into four vertical segments by three vertical lines. A fourth segment was estimated by constructing a vertical line parallel to the other lines and equal in width to the segments of the facet joint. The center of the pedicle was related to these arbitrary regions of the facet (*i.e.*, segments 1, 2, 3, and 4).

Results

The obtained data were analyzed and compared using a two-tailed student *t* test, with a *P* value less than 0.05 considered significant. Because only two female cadavers were studied, they were not compared with the previous studies.

Direct Measurements

Tables 1 and 2 show the direct measurements.)

Sagittal Diameter. In the male cadavers, the sagittal diameter decreased gradually from L1 to L4, then increased again at L5. In male specimens it ranged from 15.68 at L1–L1 to 4.79 mm at L4. The values in the female cadavers were larger at all levels, but this difference was not statistically significant ($P > 0.05$).

Sagittal Angle. The sagittal angle increased gradually from L1 (4°) to L4 (11°), then decreased again at L5 in the males. The direction was cephalad from L1 to L4, but the sagittal angle at L5 varied greatly from −8° to 15°, a negative sign indicating a caudad direction. The values in the females were lower at the upper lumber levels.

Interpedicular Distance. Interpedicular distance increased gradually from L1 to L4. In males the increase was from 32.4 to 35 cm. The values were slightly greater in females.

Transverse Diameter. The transverse diameter increased gradually from L1 to L5. The increase was from 5.95 to 16.19 mm in males. The values in females were comparatively less. In two specimens, the transverse diameter at upper lumbar levels was less than 5 cm.

Transverse Angle. The transverse angle increased gradually from L1 to L5. The increase was from 5.5° to 29.33° in males. The values in females were comparatively less at all levels except L2.

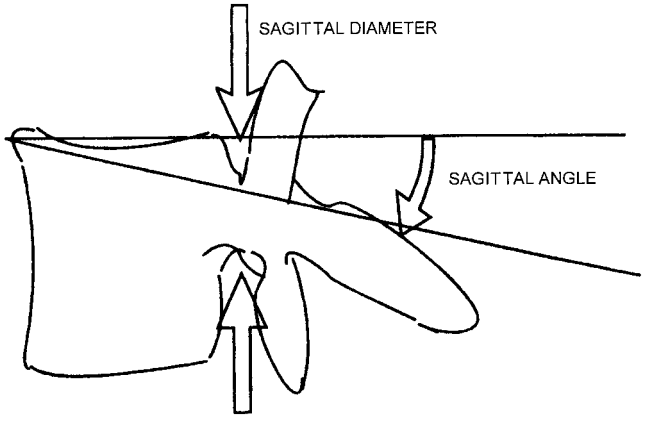


Figure 2. The method for measuring the sagittal diameter and the sagittal angle.

Table 1. Comparison of TD, TA, and CL With Previous Studies

		Study		L1	L2	L3	L4	L5
TD	Direct	CS	M(18)	7.05	7.85	9.01	11.6	16.19
		Current study	F(2)	5.95	6.5	7.55	9.25	12.75
		Kim et al[8]	M(42)	7	7.5	9.9	12.7	18.9
			F(31)	6.6	6.9	8.9	11.9	17.6
		Hou et al[7]	M(25)	7.2	7.6	9.4	10.8	12.8*
			F(15)	6.6	7.1	9	10.2	13
		Olsewski et al[18]	M(24)	9.5	9.6*	11.7*	14.7*	21.1*
			F(25)	7.7	7.9	9.6	12.5	18.4
	CT	CS	M(18)	7.5	7.59	8.65	9.81	14.64
			F(2)	5.85	6.15	7.25	8.8	13.1
		Zindrick et al[27]	(85)	8.7	8.9	10.3	12.9*	18*
TA	Direct	CS	M(18)	9	10.05	12.33	14.72	29.33
			F(2)	5.5	10	10.5	8.5	21
		Olsewski et al[18]	M(24)	7	7	8	11	17*
			F(25)	5	6	7	10	18
	CT	CS	M(18)	11	9.5	10.8	12.3*	24.3*
			F(2)	3.5	8	9.5	11.5	24.5
		Zindrick et al[27]	(85)	10.9	12	14.4	17.7*	29.8*
		Olsewski et al[18]	(51)	6	8	13	16*	29*
CL	Direct	CS	M(18)	46.01	47.56	46.9	46.31	45.87
			F(2)	41.2	45.8	47.3	46.55	41.65
		Olsewski et al[18]	M(24)	45	46.8	49	50.4	46.2
			F(25)	42.5	43.5	44.8	45.9	42.3
		Hou et al[7]	M(25)	50.7	50.3	49.7	47.9	43.9
			F(15)	47.6	48.6	46.3	46.3	43.1
	CT	CS	M(18)	46.8	49.7	49.35	49.83	46.87
			F(2)	45.8	44	45.25	44.95	46.7
		Zindrick et al[27]	(85)	44.7	45.5	44.4	40.7	33.7
		Olsewski et al[18]	(51)	45.8	49.2	51.7	55.3	49.2

* Statistically significant difference.

Chord Length. In males the maximum chord length (47.56 cm) was found at the midlumbar level (L2), decreasing gradually in upper and lower lumbar levels. The values in females were lower at all levels except L3 and L4.

Computed Tomography Scan Measurements

Tables 1 and 2 show the computed tomography scan measurements.

Table 2. Comparison of SD, SA, and IPD With Previous Studies

		Study	Level	L1	L2	L3	L4	L5
SD	Direct	CS	M(18)	15.68	15.27	15.03	14.79	15.67
			F(2)	15.7	15.7	15.4	14.7	17
		Olsewski et al[18]	M(24)	17	16	16	16.4	17.4
			F(25)	15.3	15.3	15	14.9	16.2
		Hou et al[7]	M(25)	15.9	15.4	15.3	15.3	20.5
			F(15)	15.5	14.7	14.3	14.5	18.7
		B&S	(30)	15.6	15.4	14.6	13	13.8
	Roentgenogram	CS	M(18)	16.3	15.5	15.1	15.1	14.8
			F(2)	17.5	17	16.5	17	18.5
		Olsewski et al[18]	(49)	18.2	17.2	16.9	15.6	13.8
		Zindrick et al[27]	(85)	15.4	15	14.9	14.8	14
SA	Direct	CS	M(18)	8.4	9.3	9.4	10.4	6.7
			F(2)	4	6	8	11	7
		Olsewski et al[18]	M(24)	5	6	6	6	5(−8)
			F(25)	6	5	6	7	8
	Roentgenogram	CS	M(18)	5	5.8	5.9	5.2	0.6
			F(2)	1.5	0	4	5.5	−4.5
		Olsewski et al[18]	(49)	7	6	7	8	7(30),−10(12)
		Zindrick et al[27]	(85)	2.4	1.8*	0.2*	0.2*	1.8
IPD	Direct	CS	M(18)	32.4	32.8	33.4	34.1	—
			F(2)	32.5	34.5	34	35	—
	Roentgenogram	CS	M(18)	37.3	38.5	38	37.1	—
			F(2)	37.5	38	40.5	37	—
Cortical thickness	CT	CS	Lateral	0.82	0.55	0.87	0.6	1.25
			Medial	1.07	1.15	1.15	1.37	1.32

* Statistically significant difference.

Transverse Diameter. The transverse diameter increased gradually from L1 to L5 in both genders (5.85 to 14.6 cm in the males). The values were significantly lower in the females at all levels. The CT scan values were found to be less than the direct measurements at all levels, although the difference was significant only at L4 and L5 ($P < 0.05$).

Transverse Angle. The transverse angle increased gradually from the upper levels downward in both genders, and was less in females at all levels. The CT scan values were significantly lower than the direct measurements at all levels except L2 in the males and L3 in the females.

Chord Length. The chord length was maximum in the midlumbar region and decreased in the upper and lower lumbar regions in both sexes. The CT values were greater than the direct measurements at all levels except L2–L4 in the female specimens.

Medial and Lateral Cortical Thickness. It was observed that the average pedicle cortical thickness of the inner wall was greater at all levels from L1 to L5, with a significant difference at L2.

Roentgenographic Measurements

Tables 1 and 2 show the roentgenographic measurements.

Sagittal Diameter. The sagittal diameter decreased gradually from the upper levels downward in the males. In the females, it increased again at L4 and L5. The roentgenographic values were higher than the direct measurements all levels except L5 in the males, but the difference was not significant.

Sagittal Angle. The sagittal angle increased gradually from L1 to L4 in both genders, but the roentgenographic values were significantly lower than the direct measurements at all levels, and the direction was cephalad. However, significant variation was seen at L5, ranging from −4.5° to 11° A negative sign indicates a caudad direction.

Interpedicular Distance. The interpedicular distance increased gradually from the upper levels downward to L4, then decreased again at L5 in both genders. The roentgenographic values were significantly higher than the direct measurements at all levels.

Pedicle Entrance Point

At all levels, the pedicle entrance point was at the midpoint of transverse process, except for L3 and L4 in specimen 5, and L1, L2, L3, and L5 in specimen 9, in which cases it was at the bottom of the transverse process. The pedicle entrance point was found to be in segment 3 (*i.e.*, the lateral division of the facet joint in relation to the various segments of the facet joints). At L5, the entrance point was situated in the lateral third of the facet joint itself (*i.e.*, segment 4 in all the cadavers studied).

■ Discussion

Various pedicular morphometric parameters were compared with those of previous studies in the literature in an attempt to sort out the important differences and their significance as related to pedicular screw fixation. These comparisons were performed using the two-tailed student *t* test, with *P* values less than 0.05 considered significant for all parameters.

Direct Examination

Sagittal Diameter. The sagittal diameter findings are shown in Table 2. The values for sagittal diameter showed similar trends in all the studies compared, gradually decreasing from L1 to L4, then increasing again at L5. However, the values mentioned by Olsewski et al[18] for white subjects are greater at all levels than those in the current study. The sagittal diameter in the current study closely resembled that in the Chinese study. The significant difference between males and females observed at all levels in the Olsewski et al[18] study was not seen in the current study or the Chinese study. From a practical point of view, this carries little significance for deciding the pedicular screw diameter because sagittal diameter was larger than transverse diameter at almost all levels.

Sagittal Angle. The sagittal angle findings are shown in Table 2 and Figure 4. The current findings in this context closely resembled those in the study by Olsewski et al.[18] These authors also saw caudad-directed pedicles at L5, with wide variation from −8 to 15. However, the values were slightly greater at most of the levels in the current study, although they were significant only at L4 ($P < 0.05$). The number of caudad-directed pedicles was significantly less than that mentioned by Zindrick et al.[27] Wide variation at L5 demands special care at insertion of the screw.

Transverse Angle. The transverse angle findings are shown in Table 1 and Figure 6. The trend of gradual increase from L1 to L5 observed in the current study was in accordance with the findings of Olsewski et al,[18] but the values were higher in the current study, significantly so at L5 ($P < 0.05$). This suggests that pedicles are more converging in the Indian population. The current authors agree with Cheung et al,[4] who noted that Asian pedicles have a larger pedicle inclination angle (transverse angle) from L1 to L4, as compared with that of the white population. The value for L5 was 29.3° in the current study, as compared with 17° for males in the white population. Roy et al[19,20] and Saillant[21] found a 30° inclination at L5, which is roughly equal to the findings in current study.

Interpedicular Distance. A significant difference was seen in interpedicular distance, as compared with the values reported by Roy et al.[19,20] Interpedicular distance for all lumbar levels was constant (26 mm) in the study by Roy et al.[19,20] In the current study, the distance was greatest at L4–L5 and smallest at L1–L2. A significant difference of approximately 10 mm exists between the interpedicular distance measured in the current study and that reported in the Roy et al[19,20] study at all levels. This implies that the distance between the holes of the pedicular plate should be larger for the Indian population.

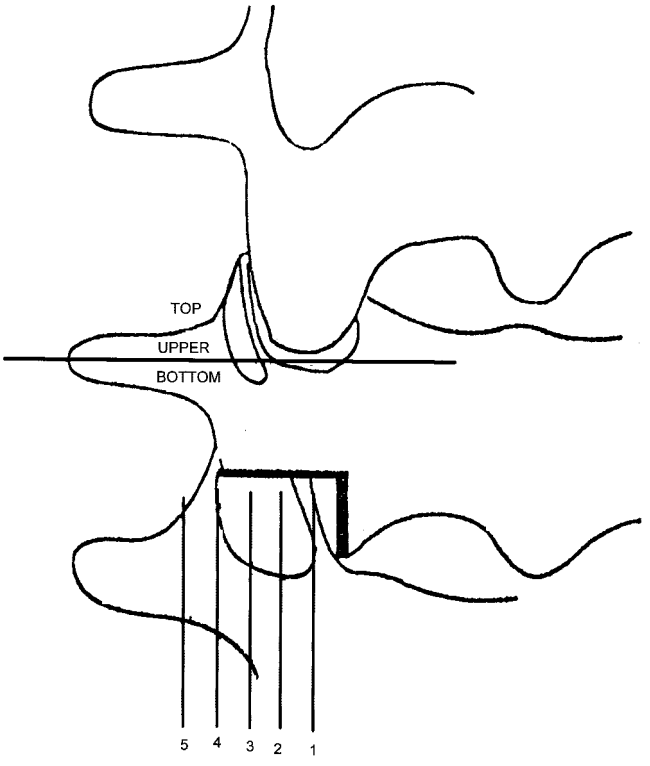


Figure 3. The method for assessing the pedicle entrance point using horizontal and vertical zones.

Transverse Diameter. The transverse diameter findings are shown in Table 1 and Figure 5. There is a gradual increase from L1 to L5 for transverse diameter in all the aforementioned studies. The values in all Asian studies are lower than those mentioned by Olsewski et al[18] for the white population. The value for L5 was 16.19 mm in the current study, as compared with is 21.1 mm for the white population studied by Olsewski et al.[18] The values for the Korean and Chinese population are 18.9 and 12.8 mm, respectively, for males. At upper lumber levels, the transverse diameter was as low as 3 mm in a few current specimens. Similar findings were noted in the Korean study also.[8] The exact cause for such a low value, apart from short stature in general, is not very well understood, although a developmental factor may be attributed. Hence, it appears that a 7-mm screw should not be used in the Indian population, especially at the upper lumber levels. Use of a 5-mm screw should be safer in the Indian population.

Chord Length. The chord length findings are shown in Table 1.[7,18] In all the studies, chord length was found to be minimum at L1 and L5, and maximum in the midlumbar levels, except in male specimens in the Chinese study. The values showed inconsistent variation among all the studies, ranging from 41.2 to 50.7 mm. In the current study, the pedicle depth was found to be greatest at L2, whereas Olsewski et al[18] found it to be greatest at L4. Hence, a shorter screw should be used at L1 and L5.

Pedicle Entrance Point. Roy et al[19,20] mentioned the pedicle entrance point at the intersection of two lines. The horizontal line passes midway through the insertion of the transverse process, and vertical line passes through the facet joint. The entrance point is thus situated 1 mm below the facet joint on a typical vertical bony crest. Hou et al[7] observed that the entrance point at lumbar levels migrates laterally from L1 to L5. At L5, it is well lateral to the lateral border of the facet joint.

In the current study, the method described by Hou et al[7] was used for the pedicle entrance point study. It was seen that the entrance point was at the midpoint of the transverse process on horizontal line and in the lateral third of the facet joint from L1 to L4, except in specimens 9 and 5.

The pedicle entrance point for L5 was well lateral to the facet joint (zone 4) in all the cadavers studied. Hence, the current findings agree with the observation of Hou et al[7] that one universal starting point would not be appropriate for all the levels.

Computed Tomography Examination

The CT findings are shown in Table 1.

Transverse Diameter and Transverse Angle. The transverse diameter and transverse angle were found to be smaller in the current study than in the White population,[18,27] although the trend from the upper levels downward was similar. The difference was significant at L4 and L5.

Chord Length. Olsewski et al[18] reported the greatest chord length at L4, as compared with L2 reported by Zindrick et al.[27] In the current study, the chord length was greatest at L4 for the males. In all the studies, the chord length at L5 consistently measured less than at the other lumbar vertebral levels. The CT values for chord length were greater than direct measurements in the current study, which accords with the findings of Olsewski et al,[18] but the difference was not consistent for the transverse diameter and the transverse angle.

Roentgenographic Examination

The roentgenographic findings are shown in Table 2.

Sagittal Diameter. The values in the current study were smaller than those mentioned by Olsewski et al,[18] but slightly larger than those mentioned by Zindrick et al,[27] although the difference was not statistically significant ($P > 0.05$). The values mentioned for the white population by Olsewski et al[18] are larger than those for the Indian population, except at L5. The roentgenographic values

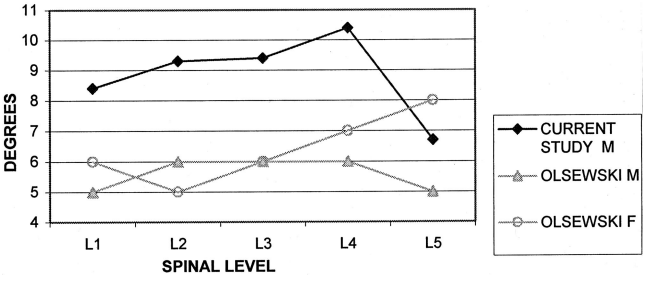


Figure 4. Comparative study of sagittal angles (direct measurements).

in the current study were slightly higher the than direct measurements, except at L5, similar to those mentioned by Olsewski et al.[18]

Sagittal Angle. For the sagittal angle, there is a significant difference between the values reported by Zindrick et al[27] and those found in the current study for L1–L4. This difference resulted mainly from the higher incidence of caudad-directed pedicles in the study by Zindrick et al.[27] However, the values of approximately 5° in the current study are in accordance with those mentioned by Olsewski et al.[18] The current finding, with those of Olsewski et al,[18] show that at L5, the sagittal angle varies widely from the cephalad to the caudad direction, denoted as positive and negative, respectively. The variation is from −10° to 8° in the Olsewski et al[18] study and from −9° to 11° in the current study.

Interpedicular Distance. Although there was an overlap of the pedicles on left and right sides, this parameter was fairly studied and showed values ranging from 37.1 to 38.5 mm for the males and from 37 to 40.5 mm for the females. These values were higher than the direct measurements, probably attributable to some degree of unavoidable overlap of the left and right pedicles.

The improved morphometric details reported in this study will help to provide the basis for stabilizing spinal implants in transpedicular screw fixation.[11] Transverse pedicle diameter showed great variation, and was almost always less than the sagittal diameter, determining the minimum screw diameter. The results suggest that a 5-mm screw would be safer at the upper lumbar levels (L1 and L2), and that a 6-mm screw would be safer at the lower lumbar levels (L3–L5). The transverse diameter values were smaller in the Indian population than in the white population. The pedicle entrance point migrated laterally for the lower lumbar levels, especially for L5. Because the medial pedicle cortex was consistently thicker than the lateral pedicle cortex, it can be safely sounded while the pedicle is probed.

Chord length was fairly constant between individuals and between vertebral levels. A screw length of 40 mm appeared to be safer for all lumbar levels. The pedicle axis was oriented anteromedially, and more obliquely at L5 (≈30°), markedly differing from that of the white population at L1–L4. The oblique orientation of the

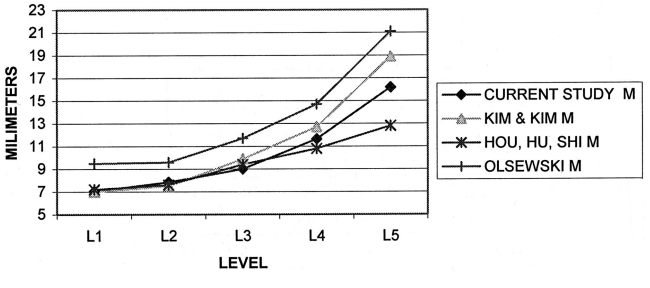


Figure 5. Comparative study of transverse diameter (direct measurements).

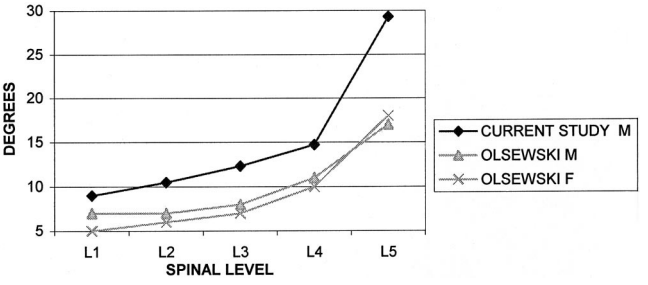


Figure 6. Comparative study of transverse angles (direct measurements).

pedicle axis allows oblique screw placement, and thus produces an interlocking or toe-nailing effect between the right and left screws in each vertebra. The values on the CT scan were slightly less (although not significant) than the direct measurements, and correspond to the findings of Krag et al[11,12] and Olsewski et al.[18] The interpedicular distance was larger in Indian population than the 26 mm observed by Roy et al.[19,20] Preoperative CT scan can give morphometric details of the pedicle, although the values are smaller than the actual (direct) measurements. The plates in the Roy et al[19,20] study had holes every 1.3 cm, and they fixed the screws at every alternate hole, the distance being 2.6 cm between the two pedicles. The plates in the study of Steffee et al[23] had a distance of 2.6 cm between the centers of two slots to accommodate the specially designed Steffee pedicular screw. These implants poorly match the interpedicular distance for the Indian population and will need modification. Moreover, for the Indian population, the diameter and length of the pedicular screw need to be substantially smaller than those mentioned in Western literature. The point of entry should be modified as mentioned. At the lower lumber levels, higher mediolateral inclination should be kept in mind (Figures 4, 5, and 6).[11,24]

■ Key Points

- Pedicular fixation is stable, versatile, and increasingly used worldwide as a preferred method of spinal fixation.
- Detailed knowledge of the pedicle morphometry is critical for proper placement of a transpedicular screw because the size and shape of the vertebral pedicle shows variation between different races.
- An attempt was made in this study to define various pedicular screw dimensions and the pedicle entrance point for the studied population.
- The pedicular screw dimensions proposed in this study may help in the development of pedicular screws for the population studied.

Acknowledgments

The authors thank Dr. Sathe, Head of the Department of Anatomy, and also Drs. Pantane and K. R. Mitra, De-

partment of Radiodiagnosis, Government Medical College, Nagpur, India.

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Address reprint requests to

Sajal R. Mitra, MS
Department of Orthopaedics
Professors Bungalow No. 4
Government Medical College
Nagpur-440003
India